

Solving the scalar wave equation using directional tight frames

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Abstract

We propose a method for solving the scalar wave equation using directional tight frames, addressing the computational and memory challenges of simulating wave propagation. The wavefield is represented in a redundant, stable directional frame, and the action of the Laplacian is decomposed into a set of pre-computed weights coupling the frame coefficients, yielding a finite-difference-like update directly in the coefficient domain. Because the frame elements have compact frequency support, these weights are sparse: only frequency-tiling neighbours couple, so the resulting operator is sparse yet essentially complete. The framework accommodates a range of sparse directional representations—wavelets, contourlets, curvelets, and wave atoms—and applies in any dimension. We validate the method on a two-dimensional, constant-velocity numerical experiment, benchmarking a curvelet propagator against a spectral reference and classical finite-difference stencils. Retaining only the neighbouring-wedge couplings reproduces the spectral reference to machine precision in relative $\mathbf{L}^2$ error and remains numerically stable, whereas neglecting the inter-wedge coupling introduces directional artefacts comparable in magnitude to finite-difference dispersion.

Keywords: wave equation, tight frames, wave propagation, curvelets

1 Introduction

The scalar wave equation $v^2 \Delta u - \partial_t^2 u = 0$ is a fundamental mathematical framework, indispensable across a broad spectrum of disciplines in which waves propagate through

a homogeneous, or nearly homogeneous, medium. The same equation governs small-amplitude pressure waves in fluids, scalar approximations of seismic P-waves, and individual Cartesian components of the electromagnetic field in homogeneous media. In room and architectural acoustics it governs the response of an enclosure filled with air; in underwater acoustics and sonar it models the propagation of pressure waves through water; and in ultrasonics and nondestructive testing it describes the interrogation of approximately uniform materials. In these settings the medium is well represented by a constant—or only slowly varying—wave speed, which is precisely the regime addressed in this work.

Even with a homogeneous medium, accurately solving the wave equation is computationally demanding: realistic three-dimensional domains, high frequencies, and long propagation times all require fine spatial and temporal sampling, and hence large memory and processing budgets [1, 2]. This cost is the main obstacle to high-fidelity simulation in the applications above, and it has motivated a range of acceleration strategies, including parallel computing, domain decomposition, and adaptive mesh refinement [3–6]. While these techniques have achieved some success in reducing computational costs, ongoing advances are needed to tackle the computational complexity of solving the wave equation across diverse applications.

Tight frames, a class of redundant signal representations, have recently garnered interest as a promising approach to mitigate the computational challenges in solving the wave equation [7, 8]. These mathematical constructs extend orthonormal bases, offering an overcomplete representation of signals while maintaining key benefits such as linear independence and stability. Utilizing tight frames enables researchers to harness the sparsity of the wavefield, which can significantly speed up the computational process and reduce the memory demands for simulating wavefields [9, 10]. Recent studies have highlighted the efficacy of tight frame-based methods across a range of wave-equation applications, including sparse reconstruction and denoising [11–13]. For instance, B. Sun et al. [8] introduced a method to solve the wave equation within the curvelet domain, showcasing the advantages of multi-scale representations for managing wave propagation challenges.

Despite the growing interest in understanding wave propagation phenomena, the application of diverse multi-scale representations—such as wavelets, curvelets, shearlets, wave atoms, and ridgelets—in solving wave equations remains relatively underexplored [8, 14–17]. These unique mathematical representations have inherent properties that could offer advantages in specific contexts, depending on the particular characteristics and complexities of the wave propagation problem. Therefore, a thorough analysis of these multi-scale approaches is essential for uncovering their potential benefits and determining appropriate contexts for their application, ultimately improving our capacity to address a broad spectrum of wave propagation issues.

When formulated within a directional frame such as curvelets, the wave equation offers several distinct advantages. Primarily, it facilitates sparsity, a property underscored by the research of Candès and Demanet [7, 18], who showed that curvelets optimally sparsify Fourier integral operators and, in particular, the solution operators of wave equations: the curvelet matrix of the wave propagator has entries that decay

faster than any polynomial, with the few significant entries clustered along shifted diagonals that track the transport of curvelet atoms along the Hamiltonian flow [18]. Additionally, by setting a threshold, insignificant curvelet coefficients can be effectively discarded, thus minimizing noise in the resulting data. This methodology selectively emphasizes certain scales and frequencies, which are vital for enhancing the clarity and accuracy of wavefield simulations.

This approach is expected to offer several further advantages, including the ability to select preferred directions for modeling wave motion, akin to ray tracing techniques [19]. For example, directions that do not significantly influence the receivers in a given region of the domain could be ignored, potentially saving computational time. Moreover, the directional properties of curvelets are expected to allow simulations to focus more precisely on the effect of specific scatterers or structures, provided the chosen curvelets’ positions and directions fall within the “light cone”, or domain of influence, associated with the structure. This picture matches the one established by Candès and Demanet [18], in which a curvelet acts as a coherent wave packet—at once wave-like and particle-like—that is transported rigidly along the Hamiltonian flow of geometrical optics. Lastly, the problem of boundaries in the simulation grid can be effectively managed by cancelling the coefficients of curvelets that are located near the boundaries and directed outward.

Building on the advantages outlined above, this study develops a method to solve wave equations across any dimensional scope using tight frame representations that are limited in frequency bandwidth. This method involves decomposing the application of the Laplacian operator to wave fields into frame vectors, each uniquely weighted by pre-computed coefficients. This approach leverages a range of sparse directional representations such as directional wavelets, contourlets, curvelets, and wave atoms, capitalizing on their distinct properties. We then validate the method on a two-dimensional, constant-velocity experiment: a curvelet propagator that retains only the couplings between neighbouring frequency-tiling wedges reproduces a spectral reference to machine precision in relative L^2 error while remaining numerically stable, and we analyse the associated offline and online computational costs. The resulting scheme offers an efficient solution to the scalar wave equation in homogeneous media, with prospective applications in room and architectural acoustics, underwater acoustics, and ultrasonics, while also improving the management of boundary conditions and the accuracy of simulations.

2 Directional tight frames

In this section, we explore the concept of directional tight frames, a method designed for efficiently and accurately representing wavefields in the context of solving the scalar wave equation. Directional tight frames are crafted from specially designed sets of functions that together form a comprehensive and stable representation of the wavefield within a defined frequency band. Employing directional tight frames enhances the compression and reconstruction of wavefield data, which in turn significantly reduces computational complexity and memory requirements.

A discrete sequence of functions $\{\varphi_\alpha\}_{\alpha \in \mathcal{L}}$ in a Hilbert space $\mathcal{H}$ is said to be a normalized tight frame of the space $\mathcal{H}$ if

$$\|u\|^2 = \sum_{\alpha \in \mathcal{L}} |\langle u, \varphi_\alpha \rangle|^2, \quad \forall u \in \mathcal{H} \quad (1)$$

Here, $\|\cdot\|$ represents the norm of the Hilbert space $\mathcal{H}$, $\langle \cdot, \cdot \rangle$ denotes the inner product in the space and $\mathcal{L}$ denotes the set of all possible parameter combinations α for the frame $\{\varphi_\alpha\}$. This definition implies that any function u in $\mathcal{H}$ can be reconstructed from the tight frame $\{\varphi_\alpha\}$:

$$u = \sum_{\alpha \in \mathcal{L}} \langle u, \varphi_\alpha \rangle \varphi_\alpha, \quad \forall u \in \mathcal{H} \quad (2)$$

$$= \sum_{\alpha \in \mathcal{L}} c_\alpha \varphi_\alpha \quad (3)$$

This relation not only confirms the completeness of the frame but also guarantees stability and robustness in representing elements within the space $\mathcal{H}$. The coefficients

$$c_\alpha = \langle u, \varphi_\alpha \rangle \quad (4)$$

represent the projection of the function u onto the frame elements φ_α . These coefficients c_α capture the contribution of each frame element φ_α to the overall function u .

An additional beneficial aspect of this framework is the Plancherel identity:

$$\langle u, u' \rangle = \sum_{\alpha \in \mathcal{L}} \langle u, \varphi_\alpha \rangle \langle \varphi_\alpha, u' \rangle, \quad \forall u, u' \in \mathcal{H} \quad (5)$$

By leveraging this identity, one can ensure the preservation of inner product relationships between functions within the Hilbert space $\mathcal{H}$. This preservation is critical for maintaining the integrity of energy and amplitude computations throughout various applications. Additionally, the Plancherel identity aids in verifying the accuracy of numerical solutions by providing a reliable means to cross-check the consistency of the transformed and original data. This verification capability is especially useful in large-scale computations where errors can propagate and amplify.

To understand the efficiency in signal representation within a specific frequency band, we present the definition of a “Directional Tight Frame”:

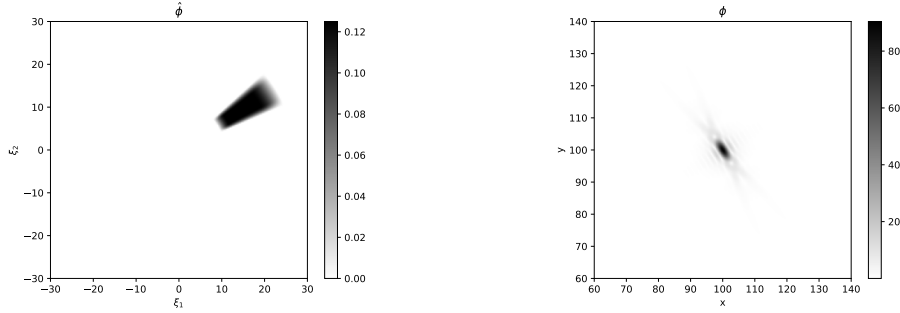
Definition 1. (*Directional Tight Frame*) Let $\{\varphi_\alpha\}_{\alpha \in \mathcal{L}}$ be a normalized tight frame for the Hilbert space $\mathcal{H} = L^2(\mathbb{R}^n)$, with α a discrete index that specifies parameters such as scale, orientation, and location. We assume that each element is band-limited: its Fourier transform $\hat{\varphi}_\alpha$ is a bump function—smooth and compactly supported in the frequency domain. The window supports overlap only with their tiling neighbours and together cover the whole frequency domain, so the frame represents all of $\mathcal{H}$ even though every individual element is band-limited. We write $\{\varphi_{\mathbf{k}}^\lambda\}_{\mathbf{k} \in \mathbb{Z}^n}^{\lambda \in \Lambda}$ for the same tight frame,

where $\mathbf{k} = (k_1, k_2, \dots, k_n)^T$ is the discrete translation index in $\mathbb{R}^n$ and Λ indexes the scale-orientation classes.

From Fourier Analysis, it is known that if $\hat{\varphi}_\alpha$ is a smooth function, then φ_α is a rapidly decreasing function. Additionally, the compact support of the window $\hat{\varphi}_\alpha$ ensures that φ_α is also smooth. Consequently, both $\hat{\varphi}_\alpha$ and φ_α reside within Schwartz space, reflecting their well-behaved nature in both the time and frequency domains.

The curvelet frame realizes this structure through a dyadic-parabolic tiling of the frequency plane (Figure 2): an isotropic low-pass disc at the coarse scale, surrounded by dyadic coronae that are partitioned into angular wedges whose number doubles every two scales. The wedges obey the parabolic scaling width $\approx \text{length}^2$ (angular width $\propto 2^{-j/2}$, radial extent $\propto 2^j$), which is the scaling that renders the action of wave propagators sparse in the resulting frame; the same construction extends to any dimension by partitioning frequency space into dyadic coronae and angular caps of opening $\sim 2^{-j/2}$ [18].

To provide a visual example of a directional tight frame, we present a curvelet both in the real space and frequency domain. In Figure 1(a), we display the curvelet window $\hat{\varphi}$ in the frequency domain. This curvelet is shown at the fourth coarsest scale and a specific angle. Figure 1(b) shows the real part of the corresponding curvelet φ , which is the Fourier transform of $\hat{\varphi}$.



(a) Curvelet window $\hat{\varphi}$ in the frequency domain.

(b) Real part of the curvelet φ in the spatial domain.

Fig. 1: Curvelet in frequency and spatial domains.

3 Scalar Wave Equation in the Directional Tight Frame Domain

The aim of this work is to solve, by using tight frames, the homogeneous scalar wave equation for a field $u : \mathbb{R}^n \times \mathbb{R} \rightarrow \mathbb{R}$,

$$v^2 \nabla^2 u(\mathbf{x}, t) - \partial_t^2 u(\mathbf{x}, t) = 0, \quad \mathbf{x} \in \mathbb{R}^n, \quad t \in \mathbb{R}. \quad (6)$$

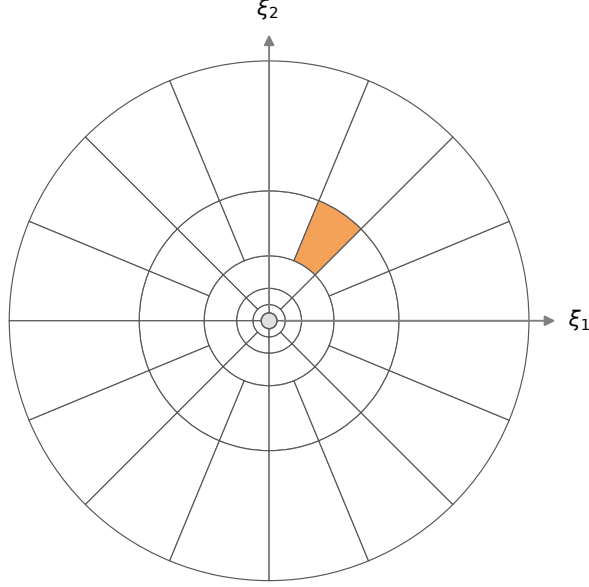


Fig. 2: Dyadic-parabolic tiling of the frequency plane by a curvelet tight frame (schematic, after [18]): an isotropic low-pass disc at the coarse scale, surrounded by dyadic coronae partitioned into angular wedges whose number doubles every two scales. Following the parabolic scaling, a wedge at scale j has angular width $\propto 2^{-j/2}$ and radial extent $\propto 2^j$. One representative wedge is shaded.

Here $\mathbf{x} = (x_1, \dots, x_n)^T$ is the spatial variable, $\nabla^2 = \sum_{i=1}^n \partial_{x_i}^2$ denotes the Laplace operator, and ∂_t^2 denotes the second derivative with respect to t . We assume a constant wave speed $v > 0$, so that the medium is homogeneous (this should not be confused with the term “homogeneous” in the wave-equation context, which implies the absence of source terms). Throughout, the field is taken to be spatially regular enough for the frame analysis that follows; specifically, for each fixed t we assume $u(\cdot, t) \in C^2(\mathbb{R}^n) \cap H^2(\mathbb{R}^n)$, the hypothesis under which Proposition 1 is stated. The initial conditions, posed as a Cauchy problem, are

$$\begin{cases} u(\mathbf{x}, 0) = u_0(\mathbf{x}), \\ \partial_t u(\mathbf{x}, 0) = g_0(\mathbf{x}), \end{cases} \quad \mathbf{x} \in \mathbb{R}^n, \quad (7)$$

where $u_0, g_0 : \mathbb{R}^n \rightarrow \mathbb{R}$ are prescribed functions.

We solve (6) by mapping it, at each fixed time t , onto the tight frame $\{\varphi_\alpha\}_{\alpha \in \mathcal{L}}$. Let $T : \mathcal{H} \rightarrow \ell^2(\mathcal{L})$ be the frame analysis operator, which assigns to a function $f \in \mathcal{H} = L^2(\mathbb{R}^n)$ the sequence of its frame coefficients,

$$(Tf)_\alpha := \langle f, \varphi_\alpha \rangle, \quad \alpha \in \mathcal{L}. \quad (8)$$

Because $\{\varphi_\alpha\}$ is a normalized tight frame, the identity (1) states precisely that $\|Tf\|_{\ell^2(\mathcal{L})} = \|f\|_{\mathcal{H}}$; in particular T is injective, so $Tf = 0$ if and only if $f = 0$. For each fixed t the wave equation (6) is an identity in $L^2(\mathbb{R}^n)$, and applying the linear operator T to it therefore loses no information: it is equivalent to the vanishing of every frame coefficient,

$$v^2 \langle \nabla^2 u, \varphi_\alpha \rangle - \langle \partial_t^2 u, \varphi_\alpha \rangle = 0, \quad \forall \alpha \in \mathcal{L}. \quad (9)$$

Define the frame coefficients of the two operators acting on the field,

$$c_\alpha^{\partial_t^2} \equiv \langle \partial_t^2 u, \varphi_\alpha \rangle, \quad c_\alpha^{\nabla^2} \equiv \langle \nabla^2 u, \varphi_\alpha \rangle, \quad (10)$$

so that (9) reads

$$v^2 c_\alpha^{\nabla^2} - c_\alpha^{\partial_t^2} = 0, \quad \forall \alpha \in \mathcal{L}. \quad (11)$$

Since the frame elements φ_α do not depend on t , the time derivative commutes with the (spatial) inner product, and the coefficients $c_\alpha^{\partial_t^2}$ follow directly from the frame coefficients $c_\alpha = \langle u, \varphi_\alpha \rangle$ of the field:

$$c_\alpha^{\partial_t^2} = \partial_t^2 \langle u, \varphi_\alpha \rangle = \partial_t^2 c_\alpha. \quad (12)$$

Computing the frame coefficients $c_\alpha^{\nabla^2}$ is a more complex and involved process than computing the time differential operator. The key tool is the symmetry of the Laplacian under the inner product, which we establish first in the easier setting of two Schwartz functions — this is enough for the matrix elements $\langle \nabla^2 \varphi_{\alpha'}, \varphi_\alpha \rangle$ between frame atoms — and then extend, with more technical work, to cover the inner product $\langle \nabla^2 u, \varphi_\alpha \rangle$ where u is the field.

Lemma 1 (Symmetry of the Laplacian on Schwartz space). *Let $\phi, \psi \in \mathcal{S}(\mathbb{R}^n; \mathbb{C})$. Then*

$$\langle \nabla^2 \phi, \psi \rangle = \langle \phi, \nabla^2 \psi \rangle.$$

Proof. Both inner products are absolutely convergent because Schwartz functions belong to every $L^p(\mathbb{R}^n)$. By Plancherel,

$$\langle \nabla^2 \phi, \psi \rangle = \langle \widehat{\nabla^2 \phi}, \widehat{\psi} \rangle = \langle -|\xi|^2 \widehat{\phi}, \widehat{\psi} \rangle = \langle \widehat{\phi}, -|\xi|^2 \widehat{\psi} \rangle = \langle \widehat{\phi}, \widehat{\nabla^2 \psi} \rangle = \langle \phi, \nabla^2 \psi \rangle,$$

where the middle equality uses that the multiplier $-|\xi|^2$ is real, so $\overline{-|\xi|^2} = -|\xi|^2$ and the factor passes from one slot of the (sesquilinear) inner product to the other. $\square$

To extend the symmetry to the case where u is the field — only assumed to satisfy $u \in C^2(\mathbb{R}^n) \cap H^2(\mathbb{R}^n)$, and in general not Schwartz — we proceed via Green's identity on balls B_R of growing radius. The Plancherel route is no longer directly available, and the obstruction is the boundary integral that arises on ∂B_R : we need it to vanish at least along some subsequence of radii.

Lemma 2 (Vanishing boundary term). *Let $u \in C^2(\mathbb{R}^n) \cap H^2(\mathbb{R}^n)$ and $\varphi \in \mathcal{S}(\mathbb{R}^n)$ (the Schwartz class), and write $B_R := \{x \in \mathbb{R}^n \mid |x| < R\}$. Then there exists a sequence of radii $R_k \rightarrow \infty$ along which the boundary term of Green's second identity vanishes:*

$$\int_{\partial B_{R_k}} (\varphi \nabla u - u \nabla \varphi) \cdot n \, dS \xrightarrow{k \rightarrow \infty} 0.$$

Proof. Set $w := \varphi \nabla u - u \nabla \varphi$, the integrand of the boundary term above. Since $\varphi \in \mathcal{S}(\mathbb{R}^n)$ we have $\varphi, \nabla \varphi \in L^2(\mathbb{R}^n)$, while $u \in H^2(\mathbb{R}^n)$ gives $u, \nabla u \in L^2(\mathbb{R}^n)$. By the Cauchy–Schwarz inequality each product is integrable,

$$\int_{\mathbb{R}^n} |\varphi \nabla u| \, dx \leq \|\varphi\|_{L^2} \|\nabla u\|_{L^2} < \infty, \quad \int_{\mathbb{R}^n} |u \nabla \varphi| \, dx \leq \|u\|_{L^2} \|\nabla \varphi\|_{L^2} < \infty,$$

hence $w \in L^1(\mathbb{R}^n)^n$. Integrating in spherical shells (the coarea formula),

$$\int_0^\infty \left(\int_{\partial B_R} |w \cdot n| \, dS \right) dR = \int_{\mathbb{R}^n} |w \cdot n| \, dx < \infty,$$

so the non-negative function $g(R) := \int_{\partial B_R} |w \cdot n| \, dS$ is integrable on $(0, \infty)$, which forces $\liminf_{R \rightarrow \infty} g(R) = 0$. Picking $R_k \rightarrow \infty$ with $g(R_k) \rightarrow 0$ and bounding the signed surface integral by $g(R_k)$ yields the claim. $\square$

Remark 1. *The subsequence is all that the next result requires: the companion volume term in Green's identity converges as a full limit, so the vanishing of the boundary term along the radii R_k already yields an exact symmetry identity for the Laplacian (Proposition 1).*

Proposition 1 (Symmetry of the Laplacian under the inner product). *Let $u \in C^2(\mathbb{R}^n) \cap H^2(\mathbb{R}^n)$ and $\varphi \in \mathcal{S}(\mathbb{R}^n)$. Then the Laplacian is symmetric with respect to the inner product:*

$$\langle \nabla^2 u, \varphi \rangle = \langle u, \nabla^2 \varphi \rangle.$$

Proof. Apply Green's second identity on the ball $B_R = \{x \in \mathbb{R}^n : |x| < R\}$, valid since $u, \varphi \in C^2$:

$$\int_{B_R} (\bar{\varphi} \nabla^2 u - u \nabla^2 \bar{\varphi}) \, dV = \int_{\partial B_R} (\bar{\varphi} \nabla u - u \nabla \bar{\varphi}) \cdot n \, dS.$$

Take the radii $R_k \rightarrow \infty$ provided by Lemma 2, applied with the test function $\bar{\varphi}$ (which is again Schwartz): the right-hand side tends to 0 as $k \rightarrow \infty$. The left-hand integrand lies in $L^1(\mathbb{R}^n)$, since $\nabla^2 u \in L^2$ (from $u \in H^2$) and $\varphi \in L^2$, while $u \in L^2$ and $\nabla^2 \varphi \in L^2$ (from $\varphi \in \mathcal{S}$), so by Cauchy–Schwarz both $\bar{\varphi} \nabla^2 u$ and $u \nabla^2 \bar{\varphi}$ are integrable; hence the left-hand side converges, as a *full* limit, to $\int_{\mathbb{R}^n} (\bar{\varphi} \nabla^2 u - u \nabla^2 \bar{\varphi}) \, dV$. Equating the two limits and using $\nabla^2 \bar{\varphi} = \overline{\nabla^2 \varphi}$,

$$\int_{\mathbb{R}^n} \bar{\varphi} \nabla^2 u \, dV - \int_{\mathbb{R}^n} u \overline{\nabla^2 \varphi} \, dV = 0, \quad \text{i.e.} \quad \langle \nabla^2 u, \varphi \rangle = \langle u, \nabla^2 \varphi \rangle.$$

□

The bumps $\widehat{\varphi}_\alpha$ defining a directional tight frame are smooth and compactly supported, so the atoms φ_α are Schwartz functions; Proposition 1 therefore applies with $\varphi = \varphi_\alpha$ for every frame element, provided the field satisfies $u \in C^2(\mathbb{R}^n) \cap H^2(\mathbb{R}^n)$.

Applying it to the definition of $c_\alpha^{\nabla^2}$ (10):

$$c_\alpha^{\nabla^2} = \langle u, \nabla^2 \varphi_\alpha \rangle$$

and from the Plancherel identity (5):

$$\begin{aligned} c_\alpha^{\nabla^2} &= \sum_{\alpha' \in \mathcal{L}} \langle u, \varphi_{\alpha'} \rangle \langle \varphi_{\alpha'}, \nabla^2 \varphi_\alpha \rangle \\ &= \sum_{\alpha' \in \mathcal{L}} c_{\alpha'} \langle \varphi_{\alpha'}, \nabla^2 \varphi_\alpha \rangle \end{aligned}$$

Since $\varphi_{\alpha'}, \varphi_\alpha \in \mathcal{S}(\mathbb{R}^n)$, Lemma 1 applies to the inner product on the right, giving

$$c_\alpha^{\nabla^2} = \sum_{\alpha' \in \mathcal{L}} c_{\alpha'} \langle \nabla^2 \varphi_{\alpha'}, \varphi_\alpha \rangle$$

Now we define the weight

$$\nu_\alpha^{\alpha'} \equiv \langle \nabla^2 \varphi_{\alpha'}, \varphi_\alpha \rangle \quad (13)$$

so that

$$c_\alpha^{\nabla^2} = \sum_{\alpha' \in \mathcal{L}} \nu_\alpha^{\alpha'} c_{\alpha'} \quad (14)$$

Corollary 1 (Hermiticity of ν). *The weights satisfy $\nu_\alpha^{\alpha'} = \overline{\nu_{\alpha'}^\alpha}$.*

Proof. Applying Lemma 1 with $\phi = \varphi_{\alpha'}$ and $\psi = \varphi_\alpha$ gives $\langle \nabla^2 \varphi_{\alpha'}, \varphi_\alpha \rangle = \langle \varphi_{\alpha'}, \nabla^2 \varphi_\alpha \rangle = \overline{\langle \nabla^2 \varphi_\alpha, \varphi_{\alpha'} \rangle}$, i.e. $\nu_\alpha^{\alpha'} = \overline{\nu_{\alpha'}^\alpha}$. □

The entries $(\nu_\alpha^{\alpha'})_{\alpha, \alpha' \in \mathcal{L}}$ thus form a Hermitian kernel representing the Laplacian in the frame $\{\varphi_\alpha\}$. This algebraic property is what underlies the empirical energy stability of the curvelet propagators reported in Section 5 (Figure 6): for a Hermitian Laplacian the standard leapfrog scheme inherits the same conservation structure as in the finite-difference setting.

Let's explore what the identity (14) signifies. The Laplace Operator acts by accounting for the influence of a point's neighborhood. In the identity (14), it aggregates all the coefficients $c_{\alpha'}$ of the field u , with each term weighted by $\nu_\alpha^{\alpha'}$. Importantly, these weights do not depend on the field u and can therefore be pre-computed. This pre-computation of weights is fundamental to the finite difference method, streamlining calculations by utilizing established coefficients. The validity of this decomposition is illustrated numerically in Section 5, where the curvelet propagator built from (14) is shown to be indistinguishable from a spectral reference once the full set of cross-block couplings is retained (see Figure 4).

In the case of curvelets, despite the potential for a vast array of index combinations (α, α') , only a limited subset actually requires consideration due to the compact support of curvelets in the frequency domain. By employing the Plancherel theorem, the weight $\nu_\alpha^{\alpha'}$ is determined within the frequency domain as follows:

$$\nu_\alpha^{\alpha'} = \langle \widehat{\nabla^2 \varphi_{\alpha'}}, \widehat{\varphi_\alpha} \rangle$$

Here, $\widehat{f}$ denotes the Fourier transform of f , and the Fourier transform of the Laplacian $\nabla^2 \varphi_\alpha$ is represented by $-|\boldsymbol{\xi}|^2 \widehat{\varphi_\alpha}$, with $\boldsymbol{\xi}$ being the Cartesian coordinates in the frequency domain. Therefore, the calculation of weights $\nu_\alpha^{\alpha'}$ only needs to be conducted within the overlapping supports of $\widehat{\varphi_\alpha}$ and $\widehat{\varphi_{\alpha'}}$. If there is no intersection between the supports of $\widehat{\varphi_\alpha}$ and $\widehat{\varphi_{\alpha'}}$, then $\nu_\alpha^{\alpha'}$ is zero. In practice, for curvelets, only coinciding or neighboring windows on the frequency tiling result in significant values of $\nu_\alpha^{\alpha'}$.

The weights exhibit translational invariance:

Proposition 2 (Translational invariance of frame inner products). *The inner product $\langle \varphi_{\mathbf{k}+\Delta\mathbf{k}}^\lambda, \varphi_{\mathbf{k}}^\lambda \rangle$ is translationally invariant, that is, does not depend on the translation index $\mathbf{k}$.*

Proof. Using the Plancherel theorem, the inner product can be transformed into the frequency domain:

$$\langle \varphi_{\mathbf{k}+\Delta\mathbf{k}}^\lambda, \varphi_{\mathbf{k}}^\lambda \rangle = \langle \widehat{\varphi_{\mathbf{k}+\Delta\mathbf{k}}}^\lambda, \widehat{\varphi_{\mathbf{k}}}^\lambda \rangle$$

Using the translation property of the Fourier Transform:

$$\langle \varphi_{\mathbf{k}+\Delta\mathbf{k}}^\lambda, \varphi_{\mathbf{k}}^\lambda \rangle = \int_{\mathbb{R}^n} e^{-i\langle \mathbf{x}_{\mathbf{k}+\Delta\mathbf{k}}^\lambda, \boldsymbol{\xi} \rangle} \widehat{\varphi_0}^\lambda(\boldsymbol{\xi}) e^{i\langle \mathbf{x}_{\mathbf{k}}^\lambda, \boldsymbol{\xi} \rangle} \overline{\widehat{\varphi_0}^\lambda(\boldsymbol{\xi})} d\boldsymbol{\xi}$$

Because the translations of a given orientation-scale class λ form a lattice, the positions depend linearly on the index, $\mathbf{x}_{\mathbf{k}}^\lambda = A_\lambda \mathbf{k}$ for a fixed non-singular generator A_λ , so that $\mathbf{x}_{\mathbf{k}+\Delta\mathbf{k}}^\lambda = \mathbf{x}_{\mathbf{k}}^\lambda + \mathbf{x}_{\Delta\mathbf{k}}^\lambda$:

$$\langle \varphi_{\mathbf{k}+\Delta\mathbf{k}}^\lambda, \varphi_{\mathbf{k}}^\lambda \rangle = \int_{\mathbb{R}^n} e^{-i\langle \mathbf{x}_{\mathbf{k}}^\lambda + \mathbf{x}_{\Delta\mathbf{k}}^\lambda, \boldsymbol{\xi} \rangle} \widehat{\varphi_0}^\lambda(\boldsymbol{\xi}) e^{i\langle \mathbf{x}_{\mathbf{k}}^\lambda, \boldsymbol{\xi} \rangle} \overline{\widehat{\varphi_0}^\lambda(\boldsymbol{\xi})} d\boldsymbol{\xi}.$$

The two modulation factors $e^{\mp i\langle \mathbf{x}_{\mathbf{k}}^\lambda, \boldsymbol{\xi} \rangle}$ — the only ones carrying the dependence on $\mathbf{k}$ — cancel, leaving

$$\begin{aligned} \langle \varphi_{\mathbf{k}+\Delta\mathbf{k}}^\lambda, \varphi_{\mathbf{k}}^\lambda \rangle &= \int_{\mathbb{R}^n} e^{-i\langle \mathbf{x}_{\Delta\mathbf{k}}^\lambda, \boldsymbol{\xi} \rangle} \widehat{\varphi_0}^\lambda(\boldsymbol{\xi}) \overline{\widehat{\varphi_0}^\lambda(\boldsymbol{\xi})} d\boldsymbol{\xi} \\ &= \langle \varphi_{\Delta\mathbf{k}}^\lambda, \varphi_{\mathbf{0}}^\lambda \rangle. \end{aligned}$$

This demonstrates that the inner product depends only on the difference $\Delta\mathbf{k}$ and is independent of the specific translation index $\mathbf{k}$ within the tight frame. $\square$

Corollary 2 (Translational invariance of Laplacian weights). *The inner product $\langle \nabla^2 \varphi_{\mathbf{k}+\Delta\mathbf{k}}^\lambda, \varphi_{\mathbf{k}}^\lambda \rangle$ likewise depends only on the difference $\Delta\mathbf{k}$ and not on the translation index $\mathbf{k}$, equalling $\langle \nabla^2 \varphi_{\Delta\mathbf{k}}^\lambda, \varphi_{\mathbf{0}}^\lambda \rangle$.*

Proof. Repeat the proof of Proposition 2 with $\widehat{\varphi}_{\mathbf{k}+\Delta\mathbf{k}}^\lambda$ replaced by $\widehat{\nabla^2\varphi}_{\mathbf{k}+\Delta\mathbf{k}}^\lambda = -|\boldsymbol{\xi}|^2 \widehat{\varphi}_{\mathbf{k}+\Delta\mathbf{k}}^\lambda$. The extra factor $-|\boldsymbol{\xi}|^2$ does not depend on $\mathbf{k}$, so the modulation factors $e^{\mp i\langle \mathbf{x}_\mathbf{k}^\lambda, \boldsymbol{\xi} \rangle}$ cancel exactly as before, leaving

$$\langle \nabla^2 \varphi_{\mathbf{k}+\Delta\mathbf{k}}^\lambda, \varphi_\mathbf{k}^\lambda \rangle = - \int_{\mathbb{R}^n} |\boldsymbol{\xi}|^2 e^{-i\langle \mathbf{x}_{\Delta\mathbf{k}}^\lambda, \boldsymbol{\xi} \rangle} |\widehat{\varphi}_0^\lambda(\boldsymbol{\xi})|^2 d\boldsymbol{\xi} = \langle \nabla^2 \varphi_{\Delta\mathbf{k}}^\lambda, \varphi_0^\lambda \rangle.$$

□

Consequently, the wave equation within the context of the tight frame $\{\varphi_\alpha\}_{\alpha \in \mathcal{L}}$ can be formulated as follows:

$$v^2 \sum_{\alpha' \in \mathcal{L}} \nu_{\alpha'}^{\alpha'} c_{\alpha'} - \partial_t^2 c_\alpha = 0 \quad (15)$$

The summation in equation (15) is integral to the discretization process inherent to the tight frame transform and, consequently, the Laplacian operator. For this approach to align with traditional methods of discretizing the Laplacian operator, at least one weight w in the summation must effectively counterbalance the others. This adjustment ensures that the equation (15) corresponds to the application of finite difference methods. We now show that, in the lattice limit, the weights obey the constraint

$$\sum_{\alpha'} \nu_{\alpha'}^{\alpha'} = 0, \quad (16)$$

mirroring the sum-zero condition typical in central difference coefficients used in traditional Laplacian operator discretizations.

Proposition 3 (Sum-zero condition for Laplacian weights). *With the lattice generator A_λ of Proposition 2 (so that $\mathbf{x}_{\Delta\mathbf{k}}^\lambda = A_\lambda \Delta\mathbf{k}$ for $\Delta\mathbf{k} \in \mathbb{Z}^n$), assume the bump $\widehat{\varphi}_0^\lambda$ is smooth and supported inside the first Brillouin zone of the reciprocal lattice $2\pi A_\lambda^{-T} \mathbb{Z}^n$ — that is, its support contains no non-zero reciprocal-lattice point. Then the weights satisfy the sum-zero condition*

$$\sum_{\Delta\mathbf{k} \in \mathbb{Z}^n} \langle \nabla^2 \varphi_{\mathbf{k}+\Delta\mathbf{k}}^\lambda, \varphi_\mathbf{k}^\lambda \rangle := \lim_{N \rightarrow \infty} \sum_{\Delta\mathbf{k} \in \mathbb{K}_N} \langle \nabla^2 \varphi_{\mathbf{k}+\Delta\mathbf{k}}^\lambda, \varphi_\mathbf{k}^\lambda \rangle = 0,$$

where $\mathbb{K}_N = \mathbb{Z}_{N_1} \times \cdots \times \mathbb{Z}_{N_n}$ and $\mathbb{Z}_N = \{-N, \dots, N\}$.

Proof. By Corollary 2 the summand depends only on $\Delta\mathbf{k}$, so the partial sum over $\mathbb{K}_N$ is

$$S_N = - \int_{\mathbb{R}^n} |\boldsymbol{\xi}|^2 D_N(\boldsymbol{\xi}) |\widehat{\varphi}_0^\lambda(\boldsymbol{\xi})|^2 d\boldsymbol{\xi}, \quad D_N(\boldsymbol{\xi}) = \sum_{\Delta\mathbf{k} \in \mathbb{K}_N} e^{-i\langle A_\lambda \Delta\mathbf{k}, \boldsymbol{\xi} \rangle}.$$

The finite sum is the multivariate Dirichlet kernel

$$D_N(\boldsymbol{\xi}) = \prod_{i=1}^n \frac{\sin((N_i + \frac{1}{2})(A_\lambda^T \boldsymbol{\xi})_i)}{\sin(\frac{1}{2}(A_\lambda^T \boldsymbol{\xi})_i)},$$

a bounded function with $D_{\mathbf{N}}(\mathbf{0}) = \prod_i (2N_i + 1)$. Rather than analyse this oscillatory integral directly, set

$$g(\boldsymbol{\xi}) = |\boldsymbol{\xi}|^2 |\hat{\varphi}_0^\lambda(\boldsymbol{\xi})|^2,$$

which is smooth and compactly supported (because $\hat{\varphi}_0^\lambda$ is), so its Fourier transform $\hat{g}(\mathbf{y}) = \int_{\mathbb{R}^n} g(\boldsymbol{\xi}) e^{-i\langle \mathbf{y}, \boldsymbol{\xi} \rangle} d\boldsymbol{\xi}$ is a Schwartz function. Interchanging the finite sum with the integral,

$$S_{\mathbf{N}} = - \sum_{\Delta \mathbf{k} \in \mathbb{K}_{\mathbf{N}}} \hat{g}(A_\lambda \Delta \mathbf{k}).$$

Letting $\mathbf{N} \rightarrow \infty$, the Poisson summation formula for the Schwartz function $\hat{g}$ on the lattice $A_\lambda \mathbb{Z}^n$ gives

$$\lim_{\mathbf{N} \rightarrow \infty} S_{\mathbf{N}} = - \sum_{\Delta \mathbf{k} \in \mathbb{Z}^n} \hat{g}(A_\lambda \Delta \mathbf{k}) = -C_\lambda \sum_{\mathbf{m} \in \mathbb{Z}^n} g(\boldsymbol{\xi}_{\mathbf{m}}), \quad \boldsymbol{\xi}_{\mathbf{m}} = 2\pi A_\lambda^{-T} \mathbf{m},$$

with $C_\lambda = (2\pi)^n / |\det A_\lambda| > 0$; that is, a sum of g over the reciprocal lattice (a Dirac comb). Each term vanishes: at $\mathbf{m} = \mathbf{0}$ the factor $|\boldsymbol{\xi}|^2$ forces $g(\mathbf{0}) = 0$, and for $\mathbf{m} \neq \mathbf{0}$ the point $\boldsymbol{\xi}_{\mathbf{m}}$ lies outside the first Brillouin zone, hence outside $\text{supp } \hat{\varphi}_0^\lambda$, so $g(\boldsymbol{\xi}_{\mathbf{m}}) = 0$. Therefore $\lim_{\mathbf{N} \rightarrow \infty} S_{\mathbf{N}} = 0$. $\square$

Remark 2. For finite $\mathbf{N}$ the partial sum equals minus the lattice tail $\sum_{\Delta \mathbf{k} \notin \mathbb{K}_{\mathbf{N}}} \hat{g}(A_\lambda \Delta \mathbf{k})$, which decays faster than any power of $\min_i N_i$ since $\hat{g}$ is Schwartz. The sum-zero condition $\sum_\alpha \nu_\alpha^{\alpha'} = 0$ thus holds exactly in the lattice limit and to high accuracy for the truncated stencils used in practice, mirroring the exact sum-zero property of central finite-difference coefficients.

The following equation encapsulates the primary result of our study on the scalar wave equation within the context of directional tight frames. It highlights how the wave components interact through a set of frame coefficients:

$$v^2 \sum_{\lambda' \in \Lambda_\lambda} \sum_{\mathbf{k}' \in \mathbb{K}} \nu_{\lambda, \mathbf{k}}^{\lambda', \mathbf{k}'} c_{\lambda', \mathbf{k}'} - \partial_t^2 c_{\lambda, \mathbf{k}} = 0$$

In this formulation, Λ_λ denotes the set of all parameters for windows that align with or are adjacent to the window $\hat{\varphi}_{\mathbf{k}}^\lambda$. This parameter set focuses on critical interactions within the tight frame, which are anticipated to influence the solution to the wave equation.

4 Finite differences

In this section, we explore the integration of finite difference methods into our approach for decomposing the scalar wave equation within the directional tight frame domain. The primary benefit of using finite differences is their efficiency in discretizing continuous differential equations. This approach allows us to approximate solutions on a discrete grid, thereby simplifying the inherently complex, continuous nature of wave propagation into a series of manageable calculations suitable for computational processing.

The coefficient-domain wave equation (15) couples $c_\alpha^{\nabla^2}(t) = \sum_{\alpha' \in \mathcal{L}} \nu_\alpha^{\alpha'} c_{\alpha'}(t)$ to the second time derivative of the frame coefficient $c_\alpha(t) := \langle u(\cdot, t), \varphi_\alpha \rangle$. In practice

the sum is restricted to the neighbour set $\Lambda_\alpha \subset \mathcal{L}$ derived in §3, retaining only the frame indices coupled to α through the Laplacian. Sampling time on a uniform grid, $c_\alpha^n := c_\alpha(n \Delta t)$, and applying the leapfrog approximation

$$\partial_t^2 c_\alpha(n \Delta t) \approx \frac{c_\alpha^{n+1} - 2c_\alpha^n + c_\alpha^{n-1}}{\Delta t^2} \quad (17)$$

in (15) yields the explicit update

$$c_\alpha^{n+1} = 2c_\alpha^n - c_\alpha^{n-1} + v^2 \Delta t^2 \sum_{\alpha' \in \Lambda_\alpha} \nu_\alpha^{\alpha'} c_{\alpha'}^n. \quad (18)$$

This discretization not only maintains the high-order accuracy characteristic of traditional finite difference methods but also adds a layer of adaptability and efficiency that surpasses conventional techniques. Numerical evidence of the stability of this scheme on a homogeneous 2D acoustic test case is reported in Section 5 (Figure 6).

To begin simulations, we apply the Cauchy initial conditions u_0 and g , effectively modeling a source impulse. This initial step is essential for initiating the dynamic evolution of the wave equation over time within our specified framework.

Additionally, the numerical stability of our simulations, especially in the n -dimensional context, depends on the dimensionless Courant number, defined as:

$$C = \Delta t \sum_{i=1}^n \frac{v}{\Delta x_i} \quad (19)$$

Here, Δx_i denotes the step sizes for each degree of freedom, emphasizing the importance of accurately calculating the Courant number at the finest resolution allowed by the selected tight frame schema. This aspect is critical for ensuring overall stability in the simulation, governed by the Courant-Friedrichs-Lewy (CFL) condition, which stipulates that $C < C_{\max}$, where $C_{\max}$ is an empirically determined upper bound.

It is important to note that updating the coefficients in our model does not require computing the transform itself. This flexibility allows for the use of transforms with perfect polar tiling, presenting a viable alternative to the pseudo-polar tiling typical of the fast curvelet transform introduced by Candès. This adaptability enhances computational efficiency and accuracy, providing a solid foundation for detailed and precise simulations of acoustic wave propagation.

5 A two-dimensional numerical illustration

To illustrate the proposed propagator we solve the homogeneous scalar wave equation (6) on a 256×256 grid with a constant velocity $v = 2000$ m/s and grid spacing $\Delta x = 10$ m. The initial condition is a radial Ricker pulse centred in the domain with peak frequency $f_0 = 30$ Hz:

$$u_0(\mathbf{x}) = \left(1 - \frac{2\pi^2 f_0^2 |\mathbf{x} - \mathbf{x}_s|^2}{v^2}\right) e^{-\frac{\pi^2 f_0^2 |\mathbf{x} - \mathbf{x}_s|^2}{v^2}}, \quad (20)$$

with zero initial velocity ($u^{-1} = u^0$ in the leapfrog scheme). The time step $\Delta t \approx 1.41$ ms satisfies the CFL condition with a safety margin of 0.4, and we run 84 steps reaching $T \approx 118.8$ ms. Boundary conditions are periodic; the wavefront does not reach the boundary in this interval.

We adopt the wrapping fast discrete curvelet transform of Candès *et al.* [7, 20] with 5 scales and 16 angles at the second-coarsest scale, yielding 190 921 complex coefficients (redundancy ≈ 2.91). The digital transform replaces the ideal polar tiling of Figure 2 by the pseudo-polar partition of Figure 3: dyadic Cartesian coronae (concentric squares) replace circular coronae, and shears with equispaced slopes $\tan \theta_\ell = \ell \cdot 2^{-\lfloor j/2 \rfloor}$ replace rotations, with the four quadrants East/North/West/South obtained by 90° rotations of the East template.

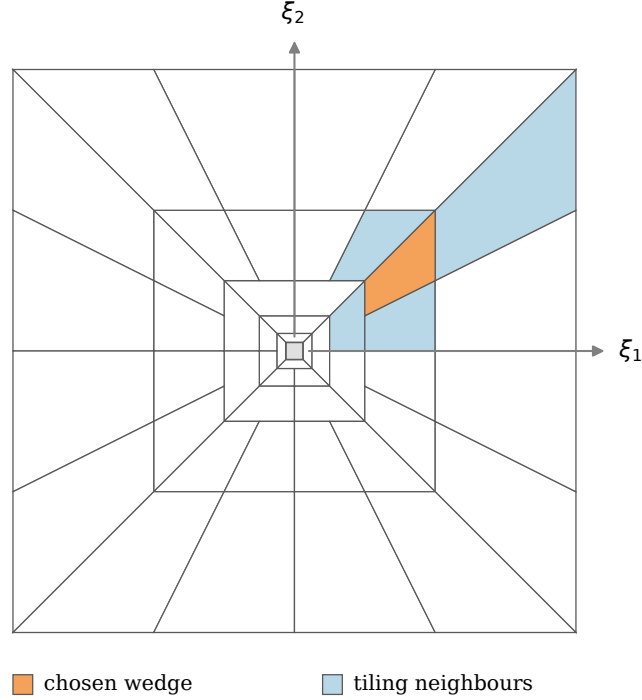


Fig. 3: Pseudo-polar (digital) curvelet tiling of the frequency plane used by the fast discrete curvelet transform [20]: dyadic Cartesian coronae partitioned into sheared trapezoidal wedges, with the four quadrants (East, North, West, South) obtained by 90° rotations of the East template. The orange wedge marks a representative atom and the light-blue wedges its frequency-tiling neighbours — the only wedges whose Fourier supports overlap with it, and therefore the only pairs for which the coupling $\nu_\alpha^{\alpha'} = \langle \nabla^2 \varphi_{\alpha'}, \varphi_\alpha \rangle$ is not numerically negligible. The block-diagonal propagator (scheme (iv) below) keeps only the orange wedge, while the neighbour-only propagator (scheme (v)) couples it to its light-blue neighbours.

The weights $\nu_{\alpha}^{\alpha'} \equiv \langle \nabla^2 \varphi_{\alpha'}, \varphi_{\alpha} \rangle$ are obtained by direct evaluation of the inner products at lattice translations $|\Delta \mathbf{k}| \leq 6$, with the Laplacian computed exactly via the spectral identity $\widehat{\nabla^2 \varphi_{\alpha}} = -|\boldsymbol{\xi}|^2 \widehat{\varphi_{\alpha}}$. Five propagators are compared:

- (i) a spectral leapfrog in physical space, used as ground truth (the Laplacian is applied exactly via the FFT);
- (ii) the classical second-order finite difference scheme with five-point stencil;
- (iii) a fourth-order finite difference scheme with the separable nine-point stencil (coefficients $(-1, 16, -30, 16, -1)/12$ along each axis), included as a higher-order classical baseline;
- (iv) the curvelet leapfrog with a *block-diagonal* restriction $\nu_{\alpha}^{\alpha'} = 0$ whenever $(j', l') \neq (j, l)$ — a deliberate simplification that retains only the orange wedge of Figure 3 and completely discards the inter-wedge couplings (the light-blue wedges in the same figure);
- (v) the curvelet leapfrog with a *neighbour-only* restriction in which we retain only the pairs $(j_t, l_t) \leftrightarrow (j_s, l_s)$ whose centred-atom inner product $\langle \nabla^2 \varphi_{j_s, l_s, \mathbf{0}}, \varphi_{j_t, l_t, \mathbf{0}} \rangle$ exceeds machine precision. This is the truly minimal cutoff: every pair of wedges that is not a frequency-tiling neighbour (Figure 3) — across non-adjacent scales or across angularly distant wedges within an adjacent scale — is dropped. Empirically, this leaves only 722 out of the $82^2 = 6724$ possible (target, source) pairs, i.e. a sparsity of 10.7%, with a mean of 8.8 active sources per target.

Schemes (iv) and (v) span the operationally relevant range of cutoffs on the pre-computed ν tensor: scheme (iv) is the most aggressive, keeping only the diagonal block and discarding every cross-wedge entry; scheme (v) is the smallest cutoff that retains accuracy, with only the entries individually above machine precision (722 active pairs). The discrepancy between the two thus isolates the contribution of the cross-block terms that scheme (iv) discards, and the agreement between scheme (v) and the spectral reference (i) confirms that all the cross-block weights of practical importance live on a sparse subset of the tiling neighbours.

In scheme (iv) the truncation of ν at $|\Delta \mathbf{k}| \leq 6$ produces small spurious positive eigenvalues that would drive an unbounded exponential growth in the leapfrog recursion; since the continuous Laplacian is negative semi-definite, we restore stability by clipping the positive part of the operator spectrum in each block, which leaves the physically meaningful modes untouched.

Figure 4 shows wavefield snapshots at $t = 0, 30, 60, 90$ and 120 ms for the five schemes. The five-point stencil reproduces the expanding ring with the well-known square-shaped dispersion artefact at later times; the fourth-order nine-point stencil reduces this dispersion substantially but does not eliminate it. The block-diagonal approximation, by contrast, exhibits clear directional (“star-like”) artefacts emanating from the source: discarding the inter-wedge coupling between adjacent (j, l) blocks discards the part of the Laplacian that leaks between overlapping curvelet wedges. The neighbour-only variant (v) is visually indistinguishable from the spectral reference at every snapshot: retaining $\nu_{\alpha}^{\alpha'}$ between the few hundred frequency-tiling neighbour pairs already captures essentially all of the operator, in agreement with the heuristic that the bumps $\widehat{\varphi}_{\alpha}$ overlap appreciably only with their tiling neighbours.

The relative L^2 error against the spectral reference is plotted in Figure 5. The neighbour-only curvelet propagator stays at machine precision throughout the simulation, while the five-point stencil reaches roughly 120% relative error by $T = 120$ ms owing to its isotropic dispersion. The fourth-order nine-point stencil cuts this error by more than half (to about 47% at $T = 120$ ms) but, being still a fixed isotropic stencil, cannot match the directional scheme. The block-diagonal approximation reaches error levels comparable to the five-point stencil with a different structural cause: angular anisotropy rather than dispersion. The coefficient-domain energy $\sum_{\alpha} |c_{\alpha}|^2$, shown in Figure 6, remains bounded for both curvelet variants, confirming the numerical stability of the leapfrog scheme in the proposed frame.

This numerical experiment supports two complementary conclusions. First, the inter-wedge coupling cannot be neglected without introducing visible angular artefacts: scheme (iv) makes this concrete, since switching it off entirely produces star-like errors comparable in magnitude to the finite-difference dispersion. Second, and most importantly for the practicality of the proposed method, the cross-block couplings that matter are confined to a sparse set of frequency-tiling neighbours: scheme (v) reproduces the spectral reference to within 10^{-15} in relative L^2 using only 722 active (target, source) pairs out of the $82^2 = 6\,724$ possible. The compact frequency support of the curvelet bumps therefore yields a sparse but essentially complete operator, and the directional structure of the tight frame is recovered without paying the cost of the full inter-block coupling. The factor of $\sim 10\times$ between the active pairs and the worst-case dense tensor sets a useful target for an efficient implementation; a systematic study of how aggressively the angular and translational supports of ν can be truncated — and of the constants in the resulting stencil — is left for future work.

5.1 Computational considerations

The economy of the proposed scheme rests on a clear separation of costs: an expensive but one-shot offline phase — the pre-computation of the weight tensor $\nu_{\alpha}^{\alpha'}$ — followed by an online phase in which the leapfrog update (15) reduces to a sparse weighted sum over a small neighbourhood in coefficient space.

For the 2D test case reported above, the offline pre-computation entailed evaluating the curvelet $\varphi_{j,l,\mathbf{0}}(\mathbf{x})$ at every (j,l) pair (82 in total, distributed as $1 + 16 + 32 + 32 + 1$ across the five scales) and the inner products $\langle \nabla^2 \varphi_{j,l,\Delta\mathbf{k}}, \varphi_{j,l,\mathbf{0}} \rangle$ for all translations $|\Delta\mathbf{k}| \leq 6$. This amounted to $82 \times 13^2 = 13\,858$ inverse-transform evaluations, requiring approximately 25 minutes on a single CPU core. The translational invariance established in Proposition 2 and Corollary 2 reduces the storage of ν from a four-index tensor to a function of the lattice difference $\Delta\mathbf{k}$, so that the block-diagonal table fits well below 1 MB on disk (complex double precision). The cross-block sparsity revealed by scheme (v) above quantifies the storage needed for the full operator: of the $82^2 = 6\,724$ candidate $(j,l) \leftrightarrow (j',l')$ pairs, only 722 carry a non-zero ν (~ 9 source wedges per target on average), so a complete table of cross-block kernels remains modest — roughly $10\times$ the block-diagonal table for the same translational truncation.

The online phase has very different cost characteristics. The full coefficient vector contains 49 577 complex numbers (a redundancy of ≈ 3 relative to the image grid), so its memory footprint is below 1 MB. Each leapfrog step is, per (15), a convolution

of each coefficient block with the corresponding ν stencil and is therefore embarrassingly parallel both across blocks and across translation indices. For the 2D test case the block-diagonal propagator runs at approximately 10 ms per time step on a single core — comparable to the cost of the five-point stencil applied directly in physical space. Scheme (v) is presented here in diagnostic form rather than in the final operational form, realised as one isolated round-trip per target wedge (about 52 s per step in the present Python prototype). This cost is dominated by repeated calls to the external FDCT binary and disappears once ν is materialised on disk: in the target implementation, each step costs the same as scheme (iv) per non-zero $(j, l) \leftrightarrow (j', l')$ block, multiplied by the ~ 9 active neighbours per target. The contrast between the diagnostic cost of (v) and the per-block convolution cost of (iv) thus highlights the practical interest of a pre-computed sparse ν : once available, it removes the need for any explicit transform inside the time loop and brings the per-step cost into the same regime as a classical finite-difference scheme, while retaining the directional structure of the underlying tight frame.

Both economies trace back to the same algebraic fact: the translational invariance of Proposition 2 and Corollary 2, which asserts that $\nu_{\mathbf{k}}^{\mathbf{k}'}$ depends only on the lattice difference $\Delta\mathbf{k} = \mathbf{k}' - \mathbf{k}$ and not on the absolute pair $(\mathbf{k}, \mathbf{k}')$. Offline, this collapses what would be a four-index tensor into a kernel indexed by $\Delta\mathbf{k}$ alone, so the inner products $\langle \nabla^2 \varphi_{j,l,\Delta\mathbf{k}}, \varphi_{j,l,\mathbf{0}} \rangle$ are assembled only against a single representative atom $\varphi_{j,l,\mathbf{0}}$ per block, rather than once per lattice position. Online, the same property turns the weighted sum in (15) into a periodic convolution of each coefficient block with its kernel, applied as a per-block Fourier multiplier — an $O(B \log B)$ operation per block of size B , in place of the $O(B^2)$ dense matrix–vector product that an unrestricted inner-product representation would otherwise demand. In this sense the proposed scheme inherits, from the lattice geometry alone, the same reuse principle that lets a fixed finite-difference stencil act identically at every grid point: a single small kernel per (j, l) , reused at every translation index both at assembly time and during time stepping.

Finally, the regularisation discussed above (spectral clipping of the small positive eigenvalues introduced by truncating ν at $|\Delta\mathbf{k}| \leq 6$) costs one FFT per block at the start of the simulation and adds no overhead to the time loop. We expect that with a larger neighbourhood R , or with a sufficient set of cross-block entries, the residual positive eigenvalues vanish and the regularisation can be dispensed with.

5.2 Relation to the optimal sparsity of wave propagators

The empirical sparsity reported above is consistent with a deeper theoretical result. Candès and Demanet [18] proved that, for a wide class of linear hyperbolic equations with smooth coefficients, the curvelet matrix of the *solution operator* $E(t)$ is optimally sparse and well organised: its entries decay faster than any negative power of their index distance, and the few significant entries cluster along shifted diagonals that correspond to the rigid transport of curvelet atoms along the Hamiltonian flow of geometrical optics. Their Corollary then bounds the truncation error: keeping only the B entries closest to each diagonal yields an operator $E_B(t)$ with $\|E(t) - E_B(t)\|_{L^2 \rightarrow L^2} \leq C_M B^{-M}$ for every $M > 0$.

It is worth being precise about how this relates to the present work. The object we discretise and truncate is the weight tensor $\nu_{\alpha}^{\alpha'} = \langle \nabla^2 \varphi_{\alpha'}, \varphi_{\alpha} \rangle$, i.e. the curvelet matrix of the *Laplacian*, a frequency multiplier $-|\xi|^2$. Its sparsity is the elementary consequence of the compact frequency support of the bumps $\widehat{\varphi}_{\alpha}$: two atoms couple only when their wedges overlap, so $\nu_{\alpha}^{\alpha'} = 0$ outside the frequency-tiling neighbours, as the experiment confirms (722 of 6 724 block pairs). The Candès–Demanet theorem, by contrast, concerns the *time-evolution* operator $E(t)$, a Fourier integral operator whose sparsity is a dynamical fact and is therefore considerably stronger.

The two are linked by the time stepping itself. Our leapfrog update (15) advances the wavefield by repeated application of the sparse single-step operator built from ν ; composing these steps produces a discrete propagator whose organisation—significant entries along diagonals shifted by the flow—is exactly the structure that Theorem 1.1 of [18] predicts for $E(t)$. This gives theoretical grounds to expect the scheme to remain both sparse and accurate as the simulation advances, and it points to a natural next step: rather than re-applying the single-step operator, one could pre-compute, once and for all, a truncated discrete propagator $E_N(t)$ over a finite time increment and exploit its diagonal-shifted sparsity directly, an avenue that [18] suggests can be realised at a cost of order $O(N(\log N)^2)$.

6 Conclusions

In this study, we have developed and numerically tested the application of directional tight frames for solving the scalar wave equation, with the aim of addressing the computational and memory challenges inherent in simulating wave propagation. The approach recasts the Laplacian as a sparse, pre-computed coupling of frame coefficients, leveraging the properties of directional tight frames to enhance computational efficiency and promote sparsity in wavefield representations.

This method combines the advantages of traditional higher-order finite difference approaches with the capabilities of directional tight frames, addressing the complexities of wave propagation problems across a wide range of applications and domains. A two-dimensional, constant-velocity numerical experiment confirms the method: a curvelet propagator that retains the couplings between neighbouring frequency-tiling wedges reproduces a spectral reference to machine precision in relative L^2 error and remains numerically stable, whereas discarding the inter-wedge coupling produces directional artefacts comparable in magnitude to finite-difference dispersion. The framework is formulated for arbitrary dimension; its validation in three dimensions and in heterogeneous media remains for future work.

One notable aspect of our methodology is that tight frames, such as curvelets, can be used within this framework, though our focus has not been limited solely to them. Applying these frames at the beginning and end of simulations offers a significant opportunity to leverage their properties for promoting sparsity, a feature that could be particularly beneficial in fields such as room and underwater acoustics.

Our findings indicate that this framework is most directly suited to settings where the medium is homogeneous, or nearly so—such as room and architectural acoustics, underwater acoustics, and ultrasonic testing—in which a constant wave speed is an

accurate model and the directional sparsity of the frame can be exploited in full. Extending the method to strongly heterogeneous media, as required by geophysical applications such as Full Waveform Inversion, would call for incorporating spatially variable velocity models and is left as a longer-term direction. There are theoretical grounds for optimism here: for media that are smooth away from isolated interfaces, the curvelet sparsity of wave propagators is expected to persist, with a curvelet striking an interface giving rise to a reflected and a refracted curvelet [18].

Our numerical experiments demonstrate the approach in two dimensions, in particular its accuracy, numerical stability, and the sparsity of the coefficient-domain operator. Further work—extending the experiments to three dimensions and heterogeneous media, and systematically studying how aggressively the weight tensor can be truncated—is needed to fully leverage the capabilities of this approach across various domains.

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Compliance with Ethical Standards

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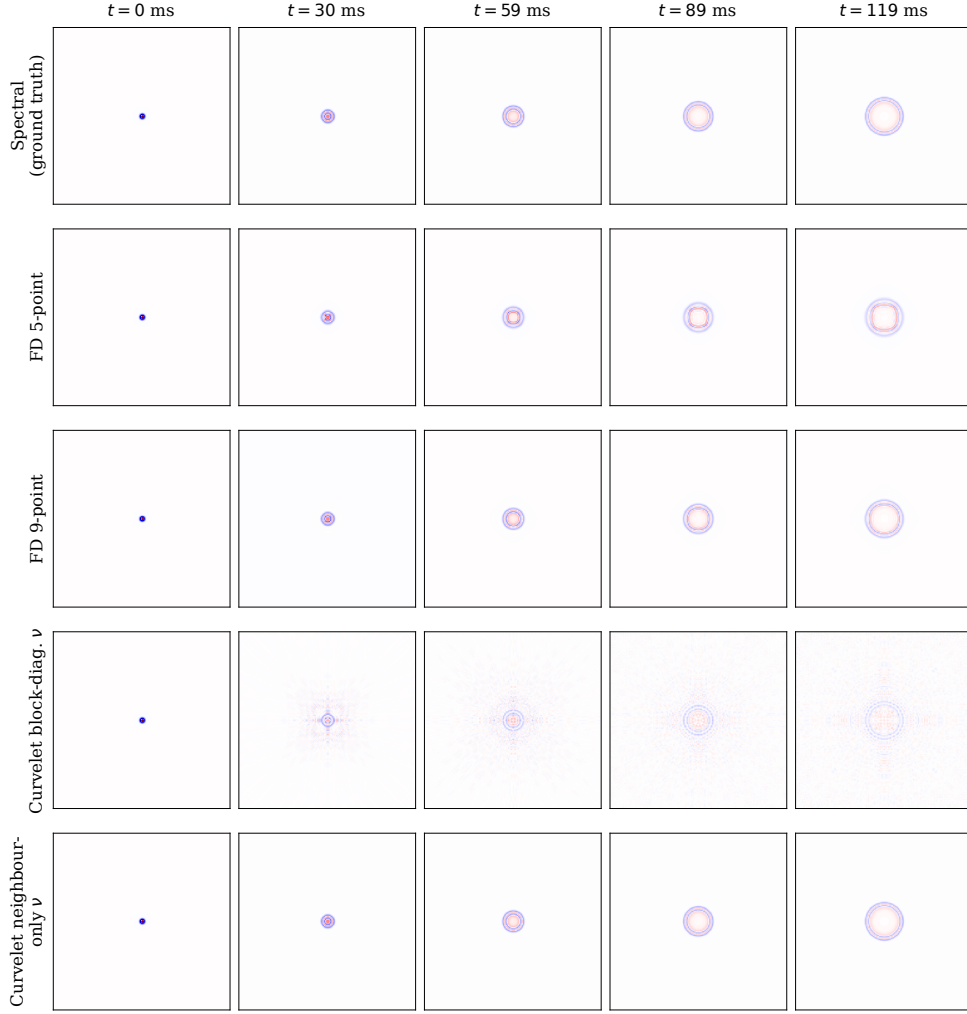


Fig. 4: Wavefield snapshots at $t = 0, 30, 60, 90, 120$ ms on a 256×256 grid with $v = 2000$ m/s, $\Delta x = 10$ m and a Ricker source ($f_0 = 30$ Hz). Rows from top to bottom: spectral leapfrog reference; classical five-point finite difference; fourth-order nine-point finite difference; curvelet leapfrog with the block-diagonal restriction $((j', l') = (j, l))$; curvelet leapfrog with the neighbour-only restriction (only frequency-tiling neighbour pairs, 722 of 6 724). The neighbour-only curvelet result is indistinguishable from the spectral reference. The block-diagonal restriction introduces visible angular artefacts whose magnitude is comparable, in L^2 , to the dispersion artefacts of the five-point stencil.

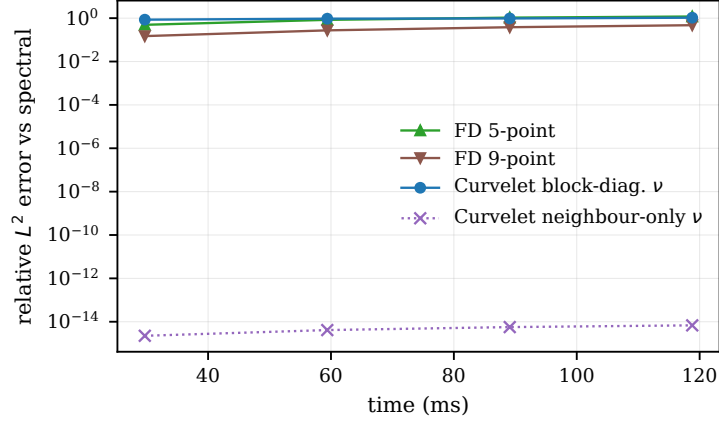


Fig. 5: Relative L^2 error of each propagator against the spectral leapfrog reference. The neighbour-only curvelet propagator stays at machine precision; the fourth-order nine-point stencil roughly halves the error of the five-point stencil but remains far from the directional scheme; the block-diagonal curvelet variant accumulates an error of comparable magnitude to the five-point stencil but with a different structural pattern (angular anisotropy rather than dispersion).

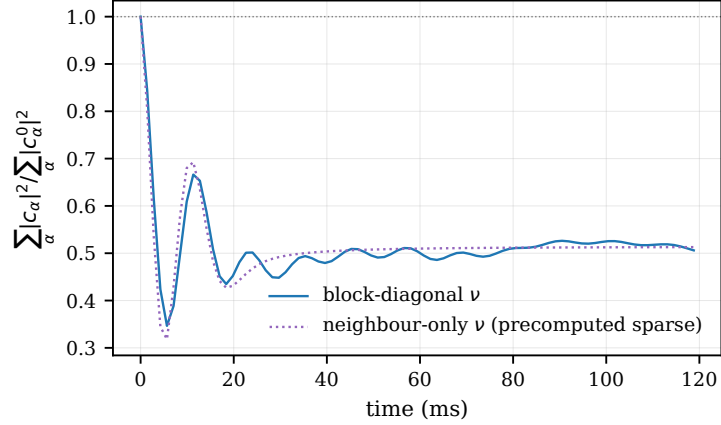


Fig. 6: Normalised coefficient-domain energy $\sum_{\alpha} |c_{\alpha}|^2 / \sum_{\alpha} |c_{\alpha}^0|^2$ as a function of time for the two curvelet propagators. Both schemes remain bounded, evidencing numerical stability of the leapfrog recursion in the directional tight frame.